

INTRODUCTORY PAPER



2017 MATHEMATICS

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

30 QUESTIONS

TIME ALLOWED: 35 MINUTES

STUDENT'S NAME:

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

1. Pete had these balloons.

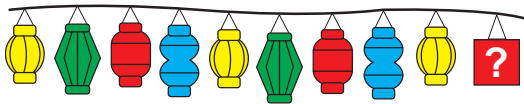


Two balloons burst.

How many were left?

- (A) 6 (B) 7 (C) 9 (D) 10

2. Anh is making a pattern with lanterns.



Which lantern should **?** be?

- (A)  (B)  (C)  (D) 

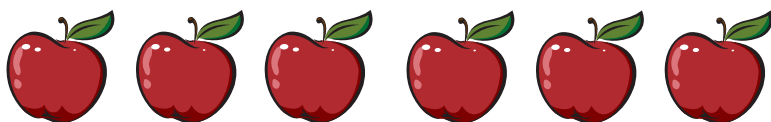
3. Cai has a vegetable garden.

He grew three rows of tomato plants with four plants in each row.

How many tomato plants did he grow?

- (A) 4 (B) 7 (C) 12 (D) 14

4. Sally had these apples.



She shared them equally between 3 people.

How many apples did each person get?

- (A) 2 (B) 3 (C) 4 (D) 6

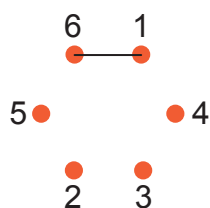
5. This table shows the number of students who ran some laps of a track. There were 14 students altogether.

Number of laps	Number of students
1	
2	
3	
4	

How many students ran **MORE** than 2 laps of the track?

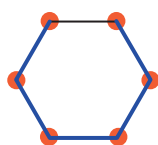
- (A) 3 (B) 4 (C) 6 (D) 8

6. Mia had this dot-to-dot picture.

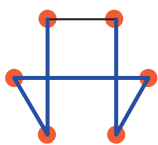


She joined the dots in order from 1 to 6.

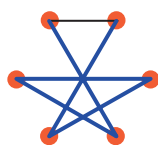
Which of these shows what Mia drew?



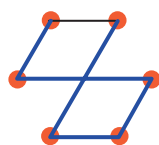
(A)



(B)



(C)



(D)

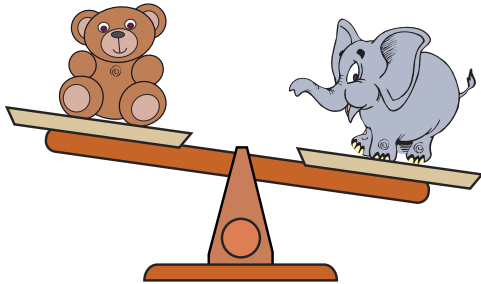
7. Lin started at 57 and counted down by the same number each time.

57, 54, 51, ?, 45, ...

What number must ? be?

- (A) 50 (B) 49 (C) 48 (D) 47

8. Simon used a balance scale to compare the masses of two toys.



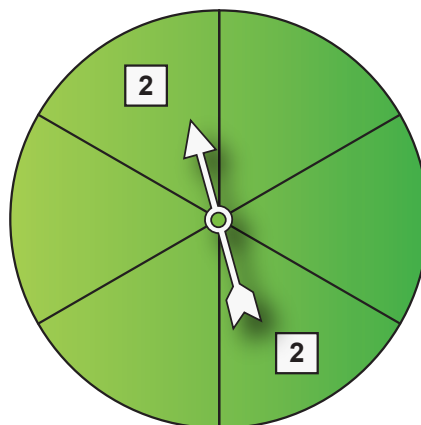
Which of these must be true?

- (A) The toys weigh the same.
(B) One toy is twice as heavy as the other.
(C) The elephant is lighter than the teddy bear.
(D) The teddy bear is lighter than the elephant.

9. Which one of these gives an answer closest to 18?

- (A) $8 + 5$ (B) $8 + 9$ (C) $9 + 5$ (D) $9 + 7$

10. Sally is making a spinner by placing 2 or 3 in each of the six parts.



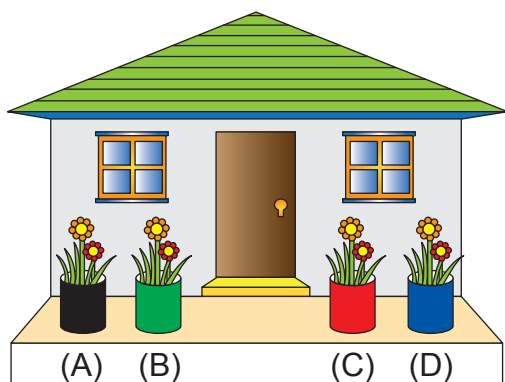
She wants the chances of the spinner landing on a 2 or on a 3 to be the same.

How many more cards with 2 does Sally need to add to her spinner?

- (A) 1 (B) 2 (C) 3 (D) 4

11. Jane faced the house and hid the key under the furthest pot to the right.

Where is the key hidden?



12. Rob's plant has two flowers.

Emma's plant has four more flowers than Rob's plant.

Which of these is Emma's plant?



(A)



(B)

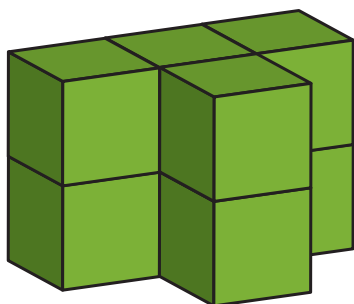


(C)



(D)

13. This solid is made from stacked cubes.



How many cubes make up this solid?

(A) 14

(B) 8

(C) 7

(D) 6

14. Which question has a number as the answer?

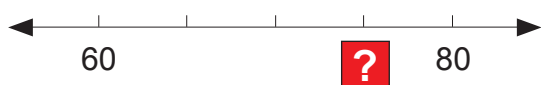
- (A) Who is the tallest in the class? (B) In what year were you born?
(C) What sport do you play? (D) Do you like eating fruit?

15. $8 + 3 = 5 + \text{?}$

What number must **?** be?

- (A) 6 (B) 7 (C) 11 (D) 16

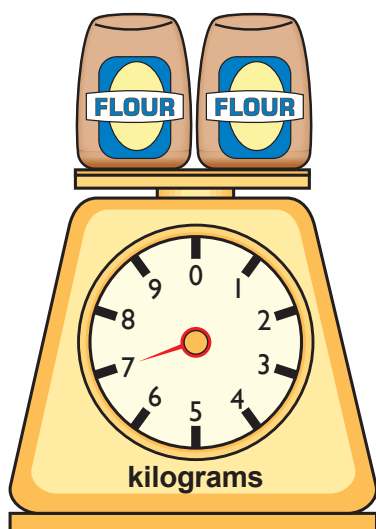
16. Leon drew this number line.



What number must **?** be?

- (A) 63 (B) 70 (C) 75 (D) 79

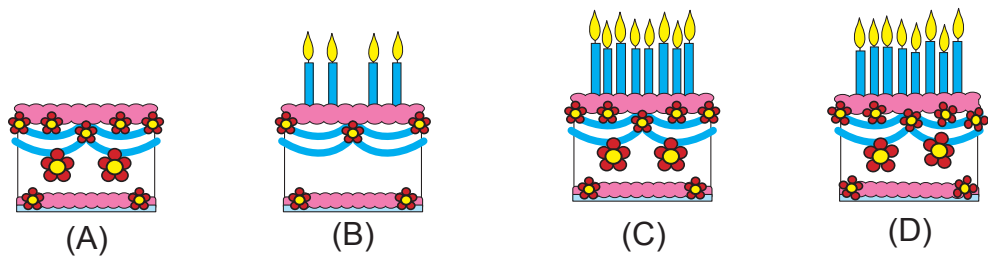
17. This picture shows the mass of 2 bags of flour.



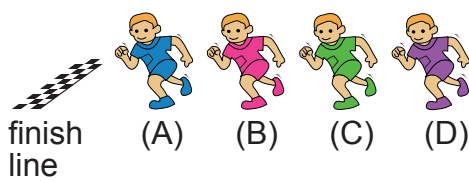
What is the mass of 1 bag of flour, in kilograms?

- (A) $3\frac{1}{2}$ (B) $4\frac{1}{2}$ (C) 7 (D) 14

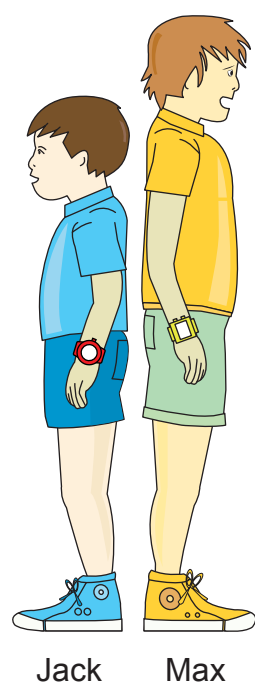
18. Which picture will **NOT** look the same after a reflection (flip)?



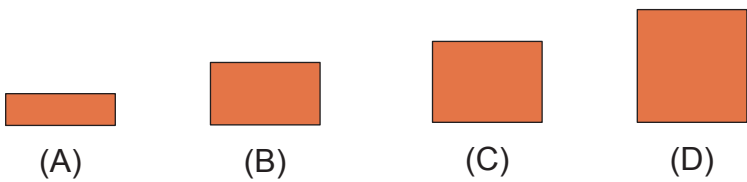
19. Four boys finished a race in the order shown.
Anish ran faster than Bob but slower than Minh. Jim came third.
Which runner is Anish?



20. Jack is shorter than Max.



Which box could Jack stand on to make the tops of their heads level?

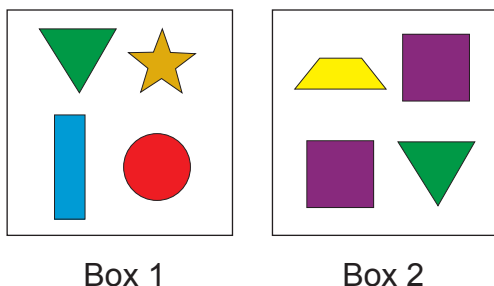


21. Jess wrote down a number.
- She said, 'If I add 4 to the tens digit and take 3 from the units, my new number will be 55'.
- What number did Jess write down?
- (A) 18 (B) 29 (C) 81 (D) 92

22. Kim stood outside a shop and read this sign painted on the window.
- Molly's**
RESTAURANT
- Kim went inside and looked at the same sign from the other side of the window.
- What did she see?

- (A)  (B) 
- (C)  (D) 

23. Aditi picked one shape from Box 1 and one shape from Box 2.



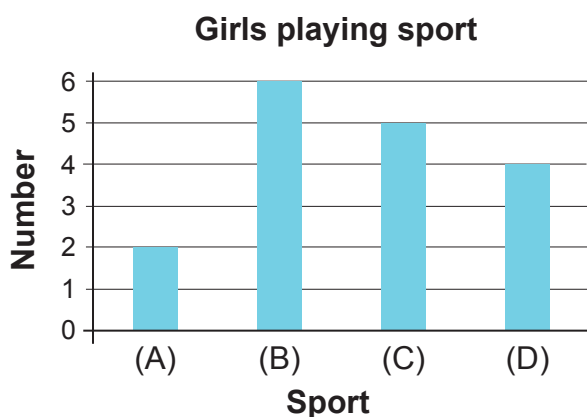
Which two shapes would be impossible for Aditi to pick?

- (A)  and 
- (B)  and 
- (C)  and 
- (D)  and 

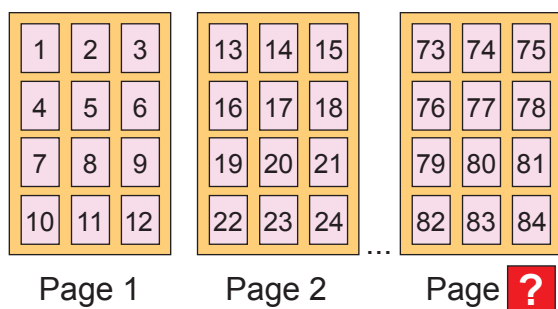
24. Each girl in Sasha's class plays only one sport.

Three more girls play tennis than play soccer.

Which column in this graph shows the girls who play tennis?



25. Lien had some cards numbered from 1 to 96. He put the cards in order then glued 12 cards to a page as shown.

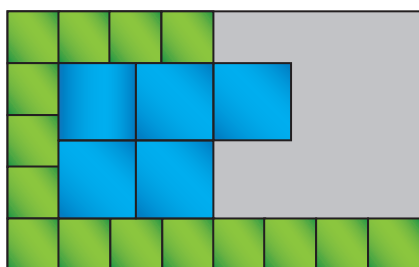


What page number must ? be?

- (A) 4 (B) 6 (C) 7 (D) 8

26. Gina started tiling a floor using large tiles and small tiles.

She used the small tiles around the outside and the large tiles in the middle, as shown.



How many tiles will Gina use altogether?

- (A) 28 (B) 30 (C) 34 (D) 40

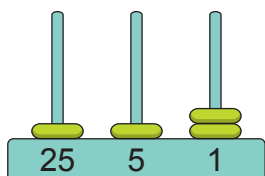
27. Jane filled this large kettle with water.



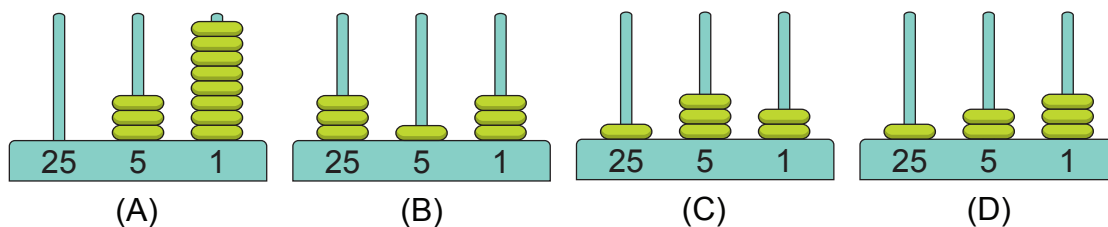
Which is closest to the number of teacups she can fill with this water?

- (A) 3 (B) 15 (C) 60 (D) 100

28. The beads on this abacus make up the number 32.



On which abacus do the beads make up the number 38?



29. Shelley made up a code.

Code	Value
	15
	40

She then wrote this line of code.



What is the value of Shelley's line of code?

- (A) 55 (B) 66 (C) 70 (D) 150

30. Kevin is solving a puzzle by placing the numbers from 1 to 9 on a grid.

	3	
1		6

The numbers in each row and in each column must add to 15.

What number must  be?

- (A) 2 (B) 4 (C) 5 (D) 7